

Java

Access specifier, packages, exception handling

Theory Notes

1. Access Specifiers

- **public** → accessible everywhere (any package, any class).
- **default (package-private)** → accessible only within the same package.
- **private** → accessible only within the same class.
- **protected** → accessible within the same package and in subclasses (even if subclass is in a different package).

2. Packages

- **Definition:** Packages are like folders that group related classes.
- **Types:**
 - **Predefined packages** → e.g., java.util, java.io.
 - **User-defined packages** → created by developers (package user;).
- **Importing:**

```
import java.util.Scanner; // predefined
```

```
import user.user;        // user-defined
```

3. Exception Handling

- **Definition:** Mechanism to handle runtime errors gracefully.
- **Keywords:**
 - **try** → block of code that may throw exception.
 - **catch** → handles exception.
 - **finally** → executes regardless of exception.
 - **throw** → used to explicitly throw an exception.
 - **throws** → declares exceptions a method may throw.
- **Common Exceptions:**
 - **ArithmeticException** → divide by zero.
 - **ArrayIndexOutOfBoundsException** → invalid array index.
 - **NullPointerException** → accessing null object.
 - **NumberFormatException** → invalid string to number conversion.

4. Best Practices

- Always catch specific exceptions before generic Exception.
- Use finally for cleanup (closing files, releasing resources).
- Avoid swallowing exceptions silently.
- Use custom exceptions for business logic errors.

Practice Tasks

Short Answer

1. What is the difference between protected and default access specifier?
2. Why should you catch specific exceptions before generic ones?
3. What is the purpose of finally block?
4. How does throws differ from throw?

Coding Exercises

1. Create a class with private, protected, default, and public fields. Try accessing them from another package.
2. Build a package student with a class Student and import it into another package.
3. Write a program that throws ArithmeticException and handles it.
4. Demonstrate multiple catch blocks for different exceptions.
5. Implement a custom exception InvalidAgeException for voting eligibility.
6. Use finally to close a file after reading, even if an exception occurs.
7. Create a subclass in a different package that accesses a protected field from its parent.

Exception.java

```
public class ExceptionHandling {  
  
    public static void main(String[] args) {  
  
        int a = 10;  
        int arr[]={1,2,3};  
  
        String s = null;  
        String num = "abc";  
  
        try {  
            // System.out.println(a/0);  
            // System.out.println(arr[5]);  
            // System.out.println(s.length());  
            System.out.println(Integer.parseInt(num));  
  
        }catch(ArithmeticException obj) {  
            System.out.println(obj.getMessage());  
        }catch(ArrayIndexOutOfBoundsException obj){  
            System.out.println(obj.getMessage());  
        }catch(NullPointerException obj){  
            System.out.println(obj.getMessage());  
        }catch(NumberFormatException obj){
```

```

        System.out.println(obj.getMessage());
    }catch(Exception e){
        System.out.println(e.getMessage());
    }

    //    Exception obj = new Exception();

    System.out.println("Below exception");
    System.out.println("Below exception");

}

}

```

Admin.java

```

package admin;

import user.user;

public class admin extends user {
    public static void main(String[] args) {

        // user obj = new user();

        // System.out.println(obj.name);

        admin obj = new admin();

        System.out.println(obj.name);
    }
}

```

User.java

```

package user;

public class user {

    // protected String name ="Sahana"; // field
    // private String name = "Sahana";
    // String name = "Sahana";
    public String name = "sahana";

    public void display(){

```

```
        System.out.println(name);
    }
}
```

publicUser.java

```
package user;

public class publicUser {

    public static void main(String[] args) {

        user obj = new user();

        System.out.println(obj.name);
    }

}
```

AccessSpecifier.java

```
import java.util.Scanner; // predefined packages

import user.user; // user-defined packages

public class AccessSpecifier {

    public static void main(String[] args) {
        System.out.println("Access specifier");

        // public , default , private, protected

        // public => access field, or method anywhere in the code
        // default => same packages has visible not different package
        // private => same class only visible
        //protected = > acces everywhere but needs extend for different
packages

        // java packages => it's like a folder
        // user obj = new user();
    }

}
```

